

The Fate of White and Black Holes – The Origin of Dark Matter

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ABSTRACT

In Series I of VG, dark matter was identified as "relics of white holes" – remnants of nodal configurations that had exhausted their emission regime. This was a static view, a snapshot of a frozen universe. With the development of Series II, we understand that the universe is alive, and "relics" are but one stage in a broader cosmic cycle. Black and white holes not only die – they transform, coalesce, disintegrate, and are reborn in other forms. What appeared to be a collection of dead objects turns out to be the very fabric from which reality continuously weaves itself. In this episode, we look beyond death, following the common destiny of white and black holes and finally revealing the profound nature of dark matter.

1. A NECESSARY RECAPITULATION: BH AND WH REGIMES IN VG

In VG, a black hole (BH) and a white hole (WH) are defined by the direction of the predominant nodal flow:

- **BH (Black Hole)** – absorption regime. The coupling arrows (A_{ij}) are oriented inward. The configuration attracts matter and energy. Flow moves from outside to inside.
- **WH (White Hole)** – emission regime. The arrows are oriented outward. The configuration ejects matter and energy. Flow moves from inside to outside.

These regimes are not eternal. They evolve, have a life, and ultimately a "death" – a transformation into something else.

2. THE LIFE CYCLE OF A MASSIVE NODAL CONFIGURATION

Any nodal configuration that reaches a critical mass (on the order of black holes) goes through the following stages:

<i>Stage</i>	<i>BH (absorption)</i>	<i>WH (emission)</i>
<i>Youth</i>	Intense absorption. Strong couplings, oriented inward.	Intense emission. Strong couplings, oriented outward.
<i>Maturity</i>	Absorption slows. The vicinity empties. Couplings weaken.	Emission slows. "Fuel" is depleted. Couplings weaken.
<i>Old Age</i>	Absorption nearly zero. Configuration enters metastable equilibrium.	Emission nearly zero. Configuration enters metastable equilibrium.
<i>Formation of the "Shell"</i>	A black shell remains – a rare fabric of the Network, with very weak couplings, transparent, but with residual mass.	A white shell remains – the same rare fabric, transparent, with residual mass.

Essential observation: In old age, the difference between BH and WH disappears. Both become **the same thing**: an inert "shell", a region of the Network where couplings are so weak that they can no longer sustain either absorption or emission.

3. WHAT ARE "SHELLS" IN VG TERMS?

A "shell" is not a particle. It is the **fabric of the Network itself, in a "frozen" state**. Imagine an old spider's web: the threads are still there, but the spider has left. It no longer catches flies, no longer vibrates, but still exists.

Properties of a shell:

- **Weak couplings** (J_{ij} near zero)
- **Null arrows** ((A_{ij}) without dominant direction)
- **Residual mass** – its presence still curves space

- **Electromagnetic transparency** – does not interact with light
- **Stability** – can exist in this state for extended cosmic periods

4. THE FINAL FATE OF SHELLS

Once formed, shells do not remain unchanged. They enter a new phase, governed by two opposing processes:

A. Coalescence – Gathering

Shells are attracted to one another, likely through a residual form of couplings that still persists. They can **merge**, forming larger structures. At the cosmic scale, these structures are visible as:

- **Dark matter filaments** – connecting galaxies
- **Halos** – surrounding galaxies
- **Groups and clusters** – massive accumulations of dark matter

Coalescence is not an active absorption, but a **passive attraction**, a reminder that all that was once separated was once one.

B. Disintegration – Dissolution

At the same time, extremely weak couplings can break over time. Nodes slowly drift apart. The fabric thins, becomes even rarer, even more transparent. Eventually, the shell may disappear completely, releasing its nodes back into the Network as free entities, ready to enter other configurations.

C. Reintegration into the Network

When the last coupling breaks, the nodes that composed the shell become again **primary nodes**. They are now free, ready to participate in new configurations, new births, new cycles. Nothing is truly lost in the Network. It only transforms.

5. SHELLS = DARK MATTER

What we observe today as **dark matter** is, in VG, precisely this immense population of shells – remnants of black and white holes that lived and died long ago.

The properties of dark matter perfectly match those of shells:

- Has **mass** (curves space, gravitationally influences galaxy motion).
- Is **transparent** (does not interact electromagnetically, does not absorb or emit light).
- Is **cold** (does not emit thermal radiation).
- Is **stable** (has existed for billions of years).
- Is **cold** in the sense that it does not interact strongly with ordinary matter.

6. CONCLUSION: THE DIFFERENCE IS ONLY IN HISTORY, NOT IN DESTINY

Aspect	White Hole (WH)	Black Hole (BH)
Youth	Emission	Absorption
Old Age	White shell	Black shell
Final Fate	Coalescence, disintegration, reintegration	Coalescence, disintegration, reintegration
Manifestation	Dark matter	Dark matter

The difference between BH and WH is only in their history. In the end, both become the same thing: the rare fabric of the Network, the dark matter that structures the universe.

7. COSMOLOGICAL IMPLICATIONS

This understanding has profound consequences:

- Dark matter is not an "exotic particle" to be sought in laboratories. It is the **frozen history of the universe itself**.
- The filamentary structure of the cosmos reflects the **coalescence** of these shells over billions of years.
- The universe is a **living, self-recycling system**: black and white holes are born, live, die, and become the matter that will nourish the next generations of stars and galaxies.